Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/08/2008
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Pharmacotherapeutics of intranasal scopolamine		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lakshmi.putcha-1@nasa.gov, jason.boyd-1@nasa.gov	**Fax:**	FY 281-244-5734
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7760
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway		
PI Address 2:	Mail Code: SK		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Boyd, Jason (USRA) Locke, James (NASA JSC)		
Grant/Contract No.:	NCC 9-58-NA00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The aims of this project entitled, “Pharmacotherapeutics of Intranasal Scopolamine,” are to (a) establish the pharmacodynamics of IN SCOP with four escalating dose levels between 0.1 and 0.4 mg, (b) examine the efficacy of two dose levels of IN SCOP for prevention and treatment of motion sickness, and (c) determine if bioavailability and pharmacodynamics of SCOP are altered in a simulated microgravity environment. Two IND phase II clinical protocols have been compiled per FDA guidelines and submitted to FDA before implementation. Standard operating procedures for scopolamine sample analysis have been validated. GLP operating procedures required in support of the project have been identified and established. An external audit by an experienced auditing agency found that record keeping and procedures relating to storage of samples, documentation, and quality assurance were outstanding and higher than their standard of excellence, the Mayo Clinic. Experimental procedure to address specific aim (a) is completed. Of the 1600 total samples collected in this protocol, analysis of plasma samples (600) is complete. Data reduction and analysis is partially completed. Experimental procedures have been partially (2/3) completed to accomplish specific aim (b). Experimental procedures to complete sample and data collection for specific aim (c) will be initiated in the second quarter of year 3 funding and targeted for completion by mid to late year 3 period. Project progress is on schedule and will be completed by the projected period.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	INSCOP can be marketed as a motion sickness medication for military personnel and for recreational air and sea travelers, once industry partnership is established for completing phase IV FDA trials.
Task Progress:	This project is designed to validate an Investigational New Drug (IND) for the treatment of space motion sickness. Three clinical trials which address study aims (a-c) have been completed. These three clinical trials generated approximately 2000 blood samples, 2000 saliva samples, and approximately 1600 urine samples. Drug concentrations have been assayed in all plasma samples. Drug concentrations have been assayed in urine and saliva of two of the three trials and the remainder will be completed by August 2008. This funded research has resulted in a new partnership with the Naval Aerospace Medical Research Association (NAMRL). NAMRL has conducted a clinical trail to compare the efficacy of four drugs, including this IND drug, in a parallel study design at suppressing motion sickness. Due to lack of perceived difference between the four different treatments the study was modified in an extension to test the comparative efficacy of the current test article, INSCOP and oral scopolamine, in a crossover study design in 16 subjects. Clinical aspects of this study were completed in early 2008. Plasma, saliva, and urine samples have been completed and data reduction and analysis is in process.
Bibliography Type:	Description: (Last Updated: 06/23/2025)
Articles in Peer-reviewed Journals	Wu L, Boyd JL, Daniels V, Wang Z, Chow DS, Putcha L. "Dose escalation pharmacokinetics of intranasal scopolamine gel formulation." J Clin Pharmacol. 2015 Feb;55(2):195-203. Epub 2014 Dec 5. https://doi.org/10.1002/jcph.391 ; PubMed PMID: 25187210 , Feb-2015
Articles in Peer-reviewed Journals	Stankovic AS, Alvarenga DL, Coleman Daniels VR, Simmons RG, Buckey JC, Putcha L. "Intranasal scopolamine for motion sickness." Aerosp Med Hum Perform. 2019 Nov;90(11):917-24. https://doi.org/10.3357/AMHP.5456.2019 ; PMID: 31666152 , Nov-2019